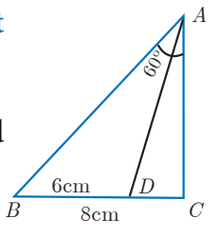
Construct of a triangle when its base, the vertical angle and the point on the base where the bisector of the vertical angle meets the base

Example 4.19 Draw a triangle ABC of base $BC = 8$ cm, $\angle A = 60^\circ$ and the bisector of $\angle A$ meets BC at D such that $BD = 6$ cm.



Rough diagram

Solution

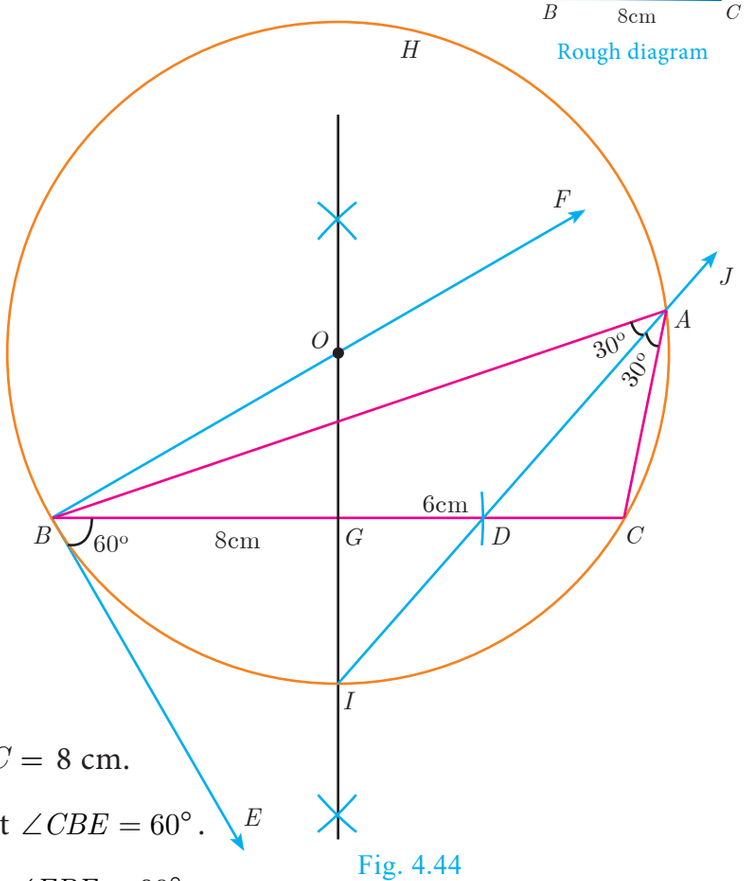


Fig. 4.44

Construction

Step 1 : Draw a line segment $BC = 8$ cm.

Step 2 : At B, draw BE such that $\angle CBE = 60^\circ$.

Step 3 : At B, draw BF such that $\angle EBF = 90^\circ$.

Step 4 : Draw the perpendicular bisector to BC, which intersects BF at O and BC at G.

Step 5 : With O as centre and OB as radius draw a circle.

Step 6 : From B, mark an arc of 6cm on BC at D.

Step 7 : The perpendicular bisector intersects the circle at I. Joint ID.

Step 8 : ID produced meets the circle at A. Now join AB and AC.

Then $\triangle ABC$ is the required triangle.

Exercise 4.2

1. In $\Delta ABC, D$ and E are points on the sides AB and AC respectively such that $DE \parallel BC$ (i) If $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15$ cm find AE.

   (ii) If $AD = 8x - 7$, $DB = 5x - 3$, $AE = 4x - 3$ and $EC = 3x - 1$, find the value of x.

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